



MALYZER

Multi-format malware analysis platform

PE · APK · Office · PDF · Scripts · JAR · ELF

TECHNICAL DOCUMENTATION

Version 1.0

R3LI4NT

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01 - About Malyzer

Defensive malware analysis platform

Malyzer is a defensive malware analysis platform with a modern graphical interface, written in C# WPF for .NET 8. It is designed for malware analysts, threat hunters and incident response professionals who need an integrated tool to examine samples and compromised hosts without juggling 10 different utilities.

■ **Defensive use only.** This tool is for security analysis and research. It does not include offensive capabilities, nor is it distributed alongside malicious payloads.

Key figures

11 FORMATS	13 MODULES	26 MITRE TECHNIQUES	10 YARA RULES
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Why Malyzer?

When an unknown sample lands in the lab, the typical workflow involves opening between 5 and 10 separate tools. Malyzer condenses that flow into a single UI with automatic triage and guided drill-down.

Need	Traditional workflow	With Malyzer
Static PE analysis	PEStudio + CFF Explorer + DIE	Built-in with scoring
APK analysis	JADX + apktool	Automatic categorization
Office maldoc	oletools + olevba	OOXML + OLE + macros
Malicious PDF	peepdf + pdfid	JavaScript + OpenAction
YARA rules	yara.exe + scripts	Embedded engine + 10 rules
MITRE ATT&CK	Manual cheatsheet	Automatic on IOCs/imports
Dynamic sandbox	Cuckoo / VMRay	Local sandbox + ETW kernel

02 · Multi-format analysis

Automatic detection of 11 file types

The dispatcher detects the file type by magic bytes and delegates to the appropriate analyzer. It supports both native binaries and container formats as well as scripts. Each analyzer feeds a common `ResultadoMultiFormato` model with categorized indicators, strings, metadata and a computed verdict.

Format	Analyzer	Specific detections
Windows PE (EXE/DLL)	AnalizadorPe	Imports · sections · entropy · packers
Android APK	AnalizadorApk	28+ dangerous permissions · DEX · native libs
Office OOXML	AnalizadorOfficeOoxml	VBA macros · external content · OLE
Office OLE	AnalizadorOfficeOle	VBA streams · equation editor
PDF	AnalizadorPdf	/JavaScript · /OpenAction · embedded files
Scripts	AnalizadorScript	PS · VBS · JS · Batch · Bash · Python
Java JAR	AnalizadorJar	Manifest · classes · imports
Linux ELF	AnalizadorPe	Headers · sections · strings · IOCs
macOS Mach-O	Generic	Strings · IOCs · hashes
Generic ZIP	Generic	Content inspection
Unknown binary	AnalizadorGenerico	Hashes · entropy · regex IOCs

Possible verdicts

Verdict	Score	Meaning
Clean	0-20	No relevant indicators
Low risk	21-40	Minor indicators · review
Suspicious	41-65	Multiple indicators · investigate
Likely malicious	66-100	Clear malware pattern

03 · Application modules

13 pages organized in 4 functional groups

Analysis

Static analysis. In-depth PE/COFF inspection based on PeNet 4.0.4 and dnlib 4.4.0. DOS/PE header, sections with entropy, imports table, IOC extraction (URLs, IPs, domains, registry, paths), packer detection, hashes (MD5, SHA-1, SHA-256, SSDeep) and automatic verdict with score 0-100.

Dynamic analysis. Local sandbox with real-time monitoring of child processes, files created/modified/deleted, registry changes and initiated network connections.

Pro tools. Memory dump via MiniDumpWriteDump, XOR brute-force deobfuscation, configuration extraction from strings and partial emulation of common cryptographic functions.

Advanced intelligence

MITRE ATT&CK Mapping. Automatic detection of 26 techniques from imports, YARA hits, IOCs and dynamic behavior. Grouping by tactic with direct links to attack.mitre.org.

Sample diff + SSDeep. Compares two samples combining 5 dimensions (SSDeep 40% + DLLs 20% + Functions 15% + YARA 15% + Sections 10%) with a 0-100 score.

Capstone decoder. Real disassembly of the .text section that detects stack strings, XOR loops, API hashing and indirect calls. Auto-detection of x86/x64.

ETW Dynamic Tracing. Replaces FileSystemWatcher with Event Tracing for Windows. Captures processes, files, registry, network and DLLs at kernel level with auto-tracking of child processes.

Threat Intelligence

Threat Intel. VirusTotal v3 + AbuseIPDB + AlienVault OTX for hashes, IPs, domains and URLs with auto-detection of the IOC type.

URL Scan. VirusTotal + URLhaus + PhishTank + local heuristics (suspicious TLDs, phishing words, literal IPs, excessive URL-encoding).

Netsniff + GeolIP. Traffic capture with SharpPcap 6.3.0. Right-click on a packet opens a modal with GeolIP, WHOIS RDAP and direct shortcuts to VirusTotal, AbuseIPDB and Shodan.

System & management

System inspector. Processes, TCP/UDP connections, autorun (registry + folders), hosts file, installed protection software, disk drives. Full context menus to suspend processes, block IPs in firewall, comment out malicious hosts.

Visualization. 7 modes: per-section entropy map, imports distribution, IOC graph, byte histogram, strings map, PE layout, process tree.

Sample repository. Local SQLite with metadata (family, tags, notes, risk level, hashes, SSDeep, MITRE techniques).

PDF export. Styled reports with QuestPDF for static analysis, system, samples, netsniff and URL scan. Automatic bilingual output.

04 · MITRE ATT&CK Mapping

26 techniques automatically detected · 9 tactics

Malyzer automatically maps each sample against the MITRE ATT&CK matrix by cross-referencing information from imports, YARA rule hits, extracted IOCs and observed dynamic behavior.

Tactic	Detected techniques
Execution	T1059.001 PowerShell · T1059.003 cmd · T1106 Native API
Persistence	T1547.001 Run keys · T1547.009 Shortcut
Defense Evasion	T1027 · T1027.002 Packing · T1055 / T1055.012 Injection · T1112 · T1140 · T1497 · T1622
Credential Access	T1003.001 LSASS · T1555.003 Browsers
Discovery	T1083 Files · T1057 Process · T1082 System · T1033 User
Collection	T1056.001 Keylogging · T1113 Screenshot
Command and Control	T1071.001 Web · T1095 Non-App Layer · T1105 Tool transfer
Impact	T1486 Encryption · T1490 Inhibit Recovery

■ Each detected technique includes a direct link to its official entry on attack.mitre.org for further reading on recommended mitigations and detections.

05 · Technical stack

Language, framework and libraries

Malyzer is built on **C# 12** and **.NET 8**, with the UI in **WPF** using WindowChrome for a modern, frameless experience.

Main libraries

Package	Version	Use
PeNet	4.0.4	Pure C# PE/COFF parsing
dnlib	4.4.0	Managed .NET assembly analysis
Gee.External.Capstone	2.3.0	Multi-architecture disassembly
Microsoft.Diagnostics.Tracing.TraceEvent	3.1.10	ETW kernel events
SharpPcap	6.3.0	L2/L3 traffic capture
PacketDotNet	1.4.7	Network packet parsing
Microsoft.Data.Sqlite	8.0.4	Local sample repository
System.Management	8.0.0	WMI queries
QuestPDF	2024.10.3	PDF generation
Newtonsoft.Json	13.0.3	JSON serialization

06 · Quick installation

Get Malyzer up and running in 5 minutes

Prerequisites

Component	Version / Detail
Operating system	Windows 10 / 11 (64 bits)
.NET Runtime	.NET 8 Desktop Runtime
Npcap	Installed with "WinPcap API-compatible" mode
Permissions	Administrator (required for ETW and memory dump)

Option 1 · Pre-compiled release

Download the `.zip` file from the latest GitHub Release, extract it and run `Malyzer.exe`.

Option 2 · Build from source

```
$ git clone https://github.com/R3LI4NT/Malyzer.git
$ cd Malyzer
$ dotnet restore
$ dotnet build -c Release
$ .\bin\Release\net8.0-windows\Malyzer.exe
```

Initial setup

Preferences are stored in `%LOCALAPPDATA%\Malyzer\config.json`. API keys are **optional** — the app works without them using only local analysis.

Service	Free tier	Use
VirusTotal	4 req/min · 500/day	Hashes, IPs, domains, URLs
AbuseIPDB	1,000 req/day	IP reputation
AlienVault OTX	no practical limit	Community pulses

07 · Antivirus warning

Why Defender may flag false positives

Malyzer triggers the same heuristics as real malware because it uses many of the same system APIs. The same thing happens with Process Hacker, x64dbg, PEStudio, the Sysinternals suite, NetCat and any other advanced analysis tool.

APIs that trigger heuristics

API / Mechanism	Legitimate use in Malyzer
OpenThread / SuspendThread	Suspending suspicious processes
MiniDumpWriteDump	Memory dump of a process
netsh advfirewall	Block malicious IPs in the firewall
SharpPcap raw capture	Network traffic capture
WMI queries	Process and hardware enumeration
Registry / hosts file	Read for inspection

Solutions

1. **Exclude the directory from Defender** (recommended for personal use):

```
PS> Add-MpPreference -ExclusionPath "C:\path\to\Malyzer"
```

2. **Report as a false positive** to Microsoft via microsoft.com/wdsi/filesubmission. They usually whitelist it within 1-2 days.

3. **Sign the executable** with an Authenticode certificate. This greatly reduces false positives but involves an annual cost.

■ **Important.** Always work with real samples in isolated VMs with snapshots and restricted networking. Never run samples on production systems or on your main host.

— End of documentation —

github.com/R3LI4NT/Malyzer